# Web Scraping — Project Report

CodeAlpha Internship | Data Analytics Track | Task 1

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## About This Task

This task is about collecting data from a website automatically using Python, instead of copying it manually. This process is called **web scraping**.

## What I Did

I built a Python script that visits a website, reads its content, and extracts useful information from it — specifically quotes, their authors, and related tags.

## How I Did It

- I used two Python libraries: **Requests** to open and read the website's content, and **BeautifulSoup** to find and extract specific pieces of information from the page (like quote text and author name).
- The website I scraped contains multiple pages of quotes. My script automatically moved from one page to the next and collected data from all of them.
- Once all the data was collected, I used **Pandas** to organize it into a clean, structured table.
- Finally, I saved this table as a **CSV file**, which can be opened in Excel or used for further analysis.

## What I Learned

- How websites are structured using HTML tags and classes.
- How to identify exactly where the required data is located on a webpage.
- How to handle multiple pages automatically (pagination) instead of doing it manually one by one.
- How to organize and save scraped data in a clean, usable format.
- The importance of being respectful while scraping (adding delays between requests, checking if a website allows scraping).

## Tools & Libraries Used

| Tool | Purpose |
|---|---|
| Python | Programming language used |
| Requests | To fetch webpage content |
| BeautifulSoup | To parse and extract data from HTML |
| Pandas | To organize and save data as CSV |

## Files in This Repository

- **web_scraper.py** — Python code used for scraping
- **scraped_quotes.csv** — Final collected data (quotes, authors, tags)

Submitted as part of the CodeAlpha Data Analytics Internship.